

Numerical Analysis PhD Exam, August 2016, Part 1

Do 4 (four) problems.

1. Define a normal matrix and prove that the following are equivalent.

- (a) A is normal.
- (b) A is unitarily diagonalizable.
- (c) $\|A\|_F = (\sum |\lambda_i|^2)^{1/2}$, where $\{\lambda_i\}$ are the eigenvalues of A counted with multiplicity.

2. Let $\kappa_2(A)$ be the two-norm condition number of the square, non-singular A .

(a) Prove that

$$\kappa_2(A) = \frac{\sigma_1}{\sigma_m},$$

where σ_1 and σ_m are the largest and smallest singular values of A , respectively.

- (b) Prove or disprove: If $A = QBQ^*$ with Q unitary, then $\kappa_2(A) = \kappa_2(B)$.
- (c) Prove or disprove: If $A = CBC^{-1}$, then $\kappa_2(A) = \kappa_2(B)$.

3. Assume $A \in \mathbb{R}^{m,n}$ with $m \geq n$, $\text{rank}(A) = n$, and $b \in \mathbb{R}^m$.

- (a) Define the least squares solution to $Ax = b$.
- (b) Derive the normal equations for the least squares problem.
- (c) Prove that $A^T A$ is invertible.
- (d) Prove that the unique solution to the least squares problem is $(A^T A)^{-1} A^T b$.
- (e) Describe how to solve the least squares problem using the QR decomposition of A .

4. This problem has two parts.

- (a) Prove that P is an orthogonal projector if and only if it is Hermitian.
- (b) Let $\{q_1, q_2, \dots, q_n\}$ be an orthonormal subset of $\mathbb{C}^m$. Show that

$$P = \sum_{i=1}^n q_i q_i^*$$

is an orthogonal projector with range equal to the span of $\{q_1, q_2, \dots, q_n\}$.

5. Assume $A \in \mathbb{R}^{m,m}$.

- (a) Prove that $\langle x, y \rangle_A = x^T A y$ is an inner product on $\mathbb{R}^m$ if and only if A is symmetric and positive definite.
- (b) Assume now that A is symmetric and positive definite. If x_* is the solution to $Ax = b$ and $\{p_1, \dots, p_m\}$ is an orthonormal basis for $\mathbb{R}^m$ with respect to $\langle \cdot, \cdot \rangle_A$ and $x_* = \sum c_i p_i$, give a formula for the c_i .